SB 79: Analyzing Transit-Oriented Development in San Francisco and Santa Clara Counties

Midterm Submission

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GitHub Repository Link: https://github.com/JoshuaDSah/CP101_FP/

Research Question

California passed Senate Bill 79 (SB 79) which went into effect on July 1, 2026. The bill sets statewide development standards and upzones land around major transit stops, attempting to increase transit-oriented housing. As SB 79 is a new policy, it is unclear which regions in California would benefit most from an increase in transit-oriented development. In an attempt to determine the most cost-effective area to develop, our group's research question is: We will estimate how SB 79 will affect urban San Francisco County and suburban Santa Clara County differently across Tier 1 and Tier 2 stops, and evaluate which development is more advantageous from transit-oriented development for future policy decisions. We will measure value by scoring Tier 1 and Tier 2 stops across urban and suburban areas, giving points according to possible housing unit increase, transit "reach" in miles, and proximity to other stations.

In this iteration of our research question, we focused on being more specific about our target metrics and overall findings we sought to discuss. Previously, our question was vague and unanswerable, but we clarified our methodologies by highlighting our impact points, pointedly answering how we would define "which development is more advantageous."

Data

For the housing data, we used a large dataset containing all parcels for each county, a dataset containing the plans for each parcel for 2023-2031, and the zones associated with SB 79 TOD. In order to restrict the SB 79 zones to only the stops within each county, we joined each set of parcels into one big parcel, and used intersects to only keep zones within the designated parcel. We were only able to download the data into xlsx format, which required us to join the geometry data found in the large parcel dataset with the information found in the planned parcels dataset. One issue we ran into was the mismatch key format between San Francisco planned parcels, and the actual current parcel dataset. The current parcel used apn keys, whereas the planned parcels used blklot and mapblklot. The keys themselves also had different formatting, and we had to use regex (learned in a previous data science class) and other methodologies to standardize the keys. We created a function to clean the keys, and applied them to the apn, blklot, and mapblklot columns.

We decided to use both blklot and mapblklot matches, since blklot corresponds to specific units, whereas mapblklot is the building footprint. After running matches to both blklot and mapblklot, we joined the two datasets, and created centroids for each parcel for ease of plotting. We then used a spatial join to restrict the centroids of the parcels to only ones located within the SB 79 zones. Finally, we filtered existing_use_description for only residential or mixed use parcels, and created a separate dataset for vacant or undeveloped parcels for future projections. We eliminated 154,547 parcel rows, or roughly 65% of the dataset during this process.

For transit route data analysis, we filtered the dataset of public transit routes in California to only transit routes that intersected with the polygon boundaries of SF and Santa Clara Counties (our counties of focus) using a spatial join. This removed 4,048 routes from the original 4,549 row dataset. I also standardized column names to be lowercase, removed whitespace from column names and entries, and converted 'objectid' (an identification number for each route treated as numeric) to a string. To analyze transit routes by census block group in these counties, we filtered California census block group

shapefiles to only census block groups in SF and Santa Clara Counties, dropping 23,753 block groups out of the original 25,607. For a further analysis into each county's block group, we subsetted the census block group data frames to create a data frame that was only SF County and a data frame that was only Santa Clara County, dropping 1,173 and 681 rows, respectively. Lastly, to determine if transit-dense block groups were already within TOD zones, we conducted a spatial join to exclude census block groups that intersected with TOD zones, dropping 216 out of the 1,854 block groups in SF and Santa Clara Counties.

Methods

For housing, using the combination of existing and planned parcel data restricted to SB 79 eligible zones, we were able to find the distribution of housing capacity for both counties. We used spatial joins to combine the .xlsx datasets to the current parcels that had geometry data, as well as restrict the parcels to the SB 79 zones. We did not need to buffer the zones manually, as there were already existing datasets for us to utilize.

We decided to calculate and visualize transit route density by census block group and TOD zone to determine where transit route density currently exists across our target counties and see how transit route density differs between urban and suburban areas. To generate transit route density values, we used a spatial join of transit route data and census block group shapefiles, and calculated how many routes intersected with each block group. After mapping these route totals to the census block group data frame, we divided these route totals by the area in square miles of each block group to get us a transit route density value. This allowed us to visualize each county's current transit route infrastructure, providing us with a baseline to compare potential TOD development benefits against. We completed a similar spatial join and route density calculation to see how many public transit routes intersected with each TOD zone, generating each zone's corresponding route density. This will help us determine later on whether TOD development is most beneficial where strong transit already exists. Lastly, we spatially joined census block group data (now attached with route density values) to exclude TOD zones, and conducted the same transit route density analysis. This was used to determine if any census block groups were transit-dense but not located in a TOD zone to be identified as potential locations for Tier 1 or Tier 2 development.

We used the NAD83 California Zone 3 (feet) CRS (EPSG:2227) for our analyses, as this projection is well-suited for mapping the Bay Area region.

Early Results

Santa Clara County only has 555 planned housing sites affected by SB 79, significantly less than the 85,711 sites in San Francisco. This is consistent with SB 79's mission to resolve the housing crisis and encourage public transit because upzoning and increased transit use is most needed in urban areas. Additionally, the most frequent increase in housing capacity ranged from 15 to 50 units. San Francisco County went from 1,262 to 1,404 units with upzoning ranging from 15 to 50 units and Santa Clara County went from 51 to 57. Surprisingly, there are very little housing capacity changes in Santa Clara County. This may indicate that building more homes isn't a priority for Santa Clara County.

Median transit route density is much higher in SF County, with a value of 123.299 compared to the route density of Santa Clara County, with a value of 18.918, indicating that public transit infrastructure is lacking in suburban Santa Clara County and could possibly benefit from the placement of

further TOD zones. Additionally, examining transit route density by tier type shows that Tier 1 zones located a half mile away from the transit stop have a higher mean transit route density (874.299) than zones located a quarter mile away from the stop (91.706). This is surprising, but indicates that there may be clusters of transit-dense zones that are currently not classified as being within 200 ft of a transit stop that potentially could become a Tier 1 zone on its own. Lastly, the median and average transit densities are much lower for both SF and Santa Clara Counties when excluding SB 79 zones, indicating that TOD zones have already taken advantage of areas with a dense amount of public transit routes.

Plan to Finish

Caleb and Josh's Responsibilities - Target Date: 11/16

- There are a few TOD zones in Santa Clara County that don't have any affected housing sites which indicates that we are likely missing important housing planning data. We'll be looking for the missing data.
- Investigate what characteristics of the Santa Clara housing sites upzoned by SB 79 make them different from the other SB 79 affected sites. What similarities do they have to the San Francisco sites upzoned by SB 79?
- Utilize metrics provided by the SB 79 plan, namely the increase in max density, to find differences in housing capacity between possible SB 79 developments and the current baseline numbers articulated by the parcels data.
- Normalize our findings, find percentiles, and associate scores with zones, tiers, and counties for interpretations

Claudia and Kianna's Responsibilities - Target Date: 11/20

- Normalize transit route density values by population.
- Examine transit route densities in census block groups containing TOD zones, seeing which tier has higher transit infrastructure and if this relationship is correlated with its county.
- Calculate transit speeds and frequencies in each zone and across our target counties to determine each zone's capacity to support more riders that could potentially result from TOD-spurred housing.
- Determine connectivity between transit stops in each TOD zone to see which tier is more connected to the transit network, and see if the type of development (urban or suburban) affects this outcome.

Our biggest risk to finishing is missing housing planning data in Santa Clara County and calculating the rest of our transit metrics. If we are unable to find the missing housing planning data, we will adjust by taking these TOD zones out of our overall analysis so these zero values will not affect the spread of our data. If we need support calculating transit metrics, we will consult the teaching team on what metrics we should prioritize and see if group members working on housing can support us in our analysis. We would like feedback on our research question to see if it needs further refinement and if it can be rephrased in a more effective way. We also want feedback on our chosen metrics to determine if we should adjust how we are calculating these metrics, or change a metric altogether. We also want to know what types of visualizations and charts would be most helpful when conveying our results and help an audience understand the significance of SB 79.